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Prathamesh Dhongade

M.Sc. Mechatronics · Autonomous Mobile Robotics

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Profile

M.Sc. Mechatronics student specializing in **Autonomous Mobile Robotics** at RWU Ravensburg-Weingarten, with professional experience in robotics systems and industrial automation. Hands-on experience with **ROS2, Nav2, SLAM, Gazebo, RViz, MoveIt2, OpenCV, YOLO, LiDAR, camera and IMU sensor fusion**, covering perception, navigation, path planning, control and hardware integration. Seeking a **Robotics Internship or Master's Thesis** in autonomous systems, mobile robotics, robot perception, navigation, computer vision or intelligent robotic systems.

Technical Skills

Robotics & Middleware	**ROS2**, ROS, Nav2, RViz, Gazebo, MoveIt2, TurtleBot3, ROS2 Nodes, Topics, Services, Actions, DDS
Navigation & SLAM	**SLAM**, LiDAR SLAM, Visual SLAM, AMCL, Localization, Path Planning, DWA, RRT, Obstacle Avoidance, ros2_control
Perception & Computer Vision	**OpenCV**, **YOLO**, PCL, Object Detection, 3D Perception, LiDAR, Camera, IMU, Sensor Fusion
Programming	**Python** (Advanced), **C++** (Intermediate), MATLAB, Linux/Ubuntu, Git, Docker
Hardware & Automation	Sensor/Actuator Commissioning, Control Cabinet Wiring, Circuit Diagrams, SolidWorks, CAN Bus, OPC-UA
Safety & Engineering	ISO 13849, Robotics System Integration, Industrial Automation, CI/CD Pipelines

Education

Hochschule Ravensburg-Weingarten University (RWU) — *2024 – Present*
M.Sc. Mechatronics Engineering / *Specialisation: Autonomous Mobile Robotics* — *Ravensburg, Germany*

Dr. DY Patil Institute of Technology, Pimpri — *2020 – 2023*
B.E. Mechanical Engineering / *Thesis: Autonomous Navigation Systems* — *Pune, India*

Guru Gobind Singh Polytechnic, Nashik — *2017 – 2020*
Diploma in Mechanical Engineering / *Focus: Robotics & Automation* — *Nashik, India*

Professional Experience

Perfect Measuring Solutions — *Jun 2023 – Dec 2023*
Robotics Systems Engineer — *Pune, India*

- Built and deployed a **ROS-based navigation system** for industrial inspection robots operating in unstructured factory environments.
- Optimised SLAM algorithms, reducing computational overhead by **45%** while maintaining mapping fidelity.
- Developed a **LiDAR + vision sensor-fusion pipeline** and authored system architecture documentation used for subsequent field deployments.
- Produced control-cabinet wiring diagrams and supported hardware commissioning of robotic control components.

ZINC-AL CAST PVT LTD. — *Nov 2020 – Nov 2021*
Automation Engineer — *Pune, India*

- Designed and integrated a **mobile robot platform** for autonomous material handling in dynamic production-line environments.
- Developed custom **path-planning algorithms** and connected perception systems to the ROS navigation stack.
- Participated in commissioning, live testing and fault diagnosis of automation systems on the factory floor.

Selected Robotics Projects

TurtleBot3 Autonomous Navigation *Ongoing*
ROS2, Nav2, Gazebo, RViz, SLAM, AMCL

- Developed an autonomous TurtleBot3 navigation pipeline covering **SLAM, localization, waypoint navigation, path planning and dynamic obstacle avoidance** using ROS2/Nav2.
- Validated navigation behaviour in Gazebo simulation and real-hardware environments.

Wall-Following Mobile Robot *Ongoing*
ROS2, Python/C++, Sensors, Gazebo, Control

- Implemented sensor-based wall-following behaviour, processing distance measurements to maintain a target wall distance and testing controller behaviour in simulation.

TIAGo Mobile Manipulator *Ongoing*
ROS2, TIAGo, Gazebo, RViz, Robot Manipulation

- Developed and tested robotic behaviours on the **TIAGo mobile manipulator**, using ROS2 interfaces and Gazebo/RViz for simulation, visualization and debugging.

Autonomous Warehouse Robot *In Development*
ROS2, SLAM, Nav2, Computer Vision, Gazebo

- Designed an autonomous warehouse-robot system integrating **SLAM, navigation, object detection, path planning and pick-and-place** for indoor material-handling scenarios.

Multi-Robot Communication *In Development*
ROS2, DDS, Multi-Robot Systems

- Developing communication between two ROS2 robots using **DDS**, focusing on sharing maps, navigation goals and robot locations for coordinated operation.

Autonomous Drone Navigation *In Development*
PX4, Gazebo, Mission Planning, Waypoint Navigation

- Developing a simulated autonomous drone-navigation workflow using **PX4 and Gazebo**, including mission planning, waypoint following and obstacle-avoidance concepts.

Reinforcement Learning for Robot Navigation *In Development*
Reinforcement Learning, Isaac Sim / Gymnasium, Python

- Developing a reinforcement-learning environment for training a simulated robot to **navigate, reach target goals and avoid obstacles** using Isaac Sim or Gymnasium.

Visual-Inertial SLAM *In Development*
Camera, IMU, EKF, Visual-Inertial Estimation

- Developing a visual-inertial localization pipeline combining **camera and IMU measurements** with Extended Kalman Filtering (EKF) for robot state estimation.

Custom AMR Platform *Ongoing*
ROS2, Nav2, AMCL, Visual Odometry, ros2_control

- Built an end-to-end autonomous mobile robot platform, combining AMCL with visual odometry; achieved a reported **92% navigation success rate** with a **15 Hz real-time control loop**.

LiDAR-based SLAM – Large-Scale Indoor Environments *Jan 2025*
ROS2, LiDAR, SLAM, Nav2

- Developed a ROS2 SLAM implementation reporting **<2 cm mapping accuracy** across **10,000 m^2**, integrated with Nav2 for autonomous navigation.

Dynamic Path Planning Optimisation *Nov 2024*
ROS2, DWA, Gazebo, Path Planning

- Extended a DWA planner for dense dynamic environments, reporting **35% computational reduction** while maintaining safety margins and testing with **20+ dynamic agents** in Gazebo.

Autonomous Bush Trimming Robot *2023*
ROS2, 3D Perception, Navigation, ISO 13849

- Developed a field-robot system incorporating **3D vegetation detection, autonomous navigation and actuation control**, with safety-system design according to ISO 13849 Performance Level requirements.

Publications & Research

- **Voice Controlled Robot Using Node-MCU** — IRJMETS, Vol. 05, Issue 12, Dec. 2023. Paper ID: IRJMETS51200060361.
- **AI Based Car Parking Allotment System** — IRJMETS, Vol. 05, Issue 12, Dec. 2023. Paper ID: IRJMETS51200059968.
- **Optimised Navigation Stack for AMRs** — RWU Technical Report, 2025.
- **Contributor – Navigation2** — ROS2 Open-Source Project, 2024.

Achievements

- **Best Robotics Implementation** — RWU Hackathon 2024.
- Published **3 ROS2 packages** and contributed to the ROS2 ecosystem.

Languages

English	Professional proficiency – IELTS 7.5
German	B2 Level – actively developing

Research Interests

Autonomous Mobile Robotics • Robot Navigation • SLAM • Robot Perception • Computer Vision • Object Detection • ROS2 • Motion Planning • Sensor Fusion • Intelligent Robotic Systems

Seeking Robotics Internship / Master's Thesis Opportunity